1. Determine the value of the definite integral $\int_1^e \left(\frac{x+x^2+1}{x}\right) dx$.

2. Why can't the definite integral $\int_0^e \left(\frac{x+x^2+1}{x}\right) dx$ be computed based on our current knowledge?

3. Consider the definite integral $\int_2^4 (x+1)\, dx$.

a) Compute the value of the integral using geometry.
b) Compute the value of the integral using the FTCP2.
c) Compute the value of integral using the limit definition of the definite integral.

4. Consider the definite integral $\int_0^2 (x^2-1)\, dx$.

a) Compute the value of the integral using the FTCP2.
b) Compute the value of integral using the limit definition of the definite integral.

5. Evaluate each of the following integrals.

a) $\int_0^1 (u+2)(u-3)\, du$

b) $\int_{\pi/2}^{\pi/4} \cot^2(x)\, dx$

c) $\int_{-2}^1 (x^{-4}+1)\, dx$

d) $\int_0^1 (t^2+t+1)^2\, dt$

6. The function $H(x)$ is defined on $\left[0, \frac{\pi}{2}\right]$ by $H(x) = \int_{\cos(x)}^{\sin(x)} \sqrt{1-t^2}\,dt$.

a) Show that $H'(x) = 1$.

b) Compute the value of H at $\frac{\pi}{4}$.

c) Use the results of parts (a) and (b) to determine $H(0.2)$.

7. Compute the following limit $\lim_{h \to 0} \frac{1}{h} \int_{2}^{2+h} \sin\left(x^2\right) dx$. *Hint: Consider how rewriting this limit using properties of integrals can be helpful.*